Oxford Brookes University
Oxford OX3 0BP, United Kingdom
jcorneli@brookes.ac.uk

July 26, 2026

Editors-in-Chief
Artificial Life
MIT Press

Dear Editors-in-Chief,

Please consider the manuscript "A New Family of Operators on Cellular Automata" for publication as an Article in *Artificial Life*.

The manuscript fits the journal's scope by introducing and characterising a finite family of artificial life-like systems in which cellular-automaton rules are themselves spatially distributed, rewritten objects. It combines an exact classification of all $8! = 40{,}320$ permutation-propagators with an exhaustive computational census of their coupled genotype/phenotype dynamics. The central result is that fixed rules exist exactly for all-even cycle structures, occur with a closed-form multiplicity, and are necessarily balanced at Langton activity $\lambda = 1/2$. The empirical component then distinguishes this fixed-point structure from finite-run dynamical persistence and explores temporal and phenotype-coupled constructions that sustain structure.

The work should be of interest to readers studying cellular automata, self-modifying systems, rule evolution, and edge-of-chaos hypotheses. Its exact results are machine-checked in Lean, while the numerical data, implementation, tests, and complete figure-reproduction workflow are publicly available at the immutable commits cited in the manuscript.

This manuscript is substantially unique, has not been published in this form, and is not under consideration elsewhere. A 2015 arXiv paper by Corneli and Maclean, cited and discussed in the manuscript, reported the historical experiment from which the present operator family was reconstructed; the submitted Article supplies a new definition, exhaustive classification, formal proofs, full census, and new empirical results.

In accordance with the journal's generative-AI policy, the manuscript includes a prominent disclosure. Anthropic Claude Opus 4.8 and OpenAI Codex coding agents assisted with manuscript editing, software implementation, test construction, and adversarial checks in a human-supervised workflow. I reviewed the resulting prose and code, reran the reported computations and formal checks, and accept full responsibility for the submission.

Thank you for considering the manuscript.

Sincerely,

Joseph Corneli